Solar System Canonical Body Reference — Four-Body Parameterization

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PAPER_186: Solar System Canonical Body Reference — Four-Body Parameterization

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$$F_U(r,t) = \sum_{i=1}^{4} U_{gi} + U_m + U_A - U_b_i, \quad \kappa = 5.0\times10^{-4}\,\text{day}^{-1},\; [SSq] = 0.57$$

$$
P_\text{sw}^\text{UQFF} = \tfrac{1}{2}\rho_text{sw}v_\text{sw}^2\bigl(1 + [SSq]\cdot\exp(-\kappa,r/v_\text{sw})\bigr), \quad \kappa = 5.0\times10^{-4}\,\text{day}^{-1}
$$

Abstract

This paper documents the canonical parameter set for the four Solar System bodies encoded as CelestialBody struct instances in the CoAnQi codebase: the Sun, Earth, Jupiter, and Neptune. Each body is fully specified by twelve UQFF parameters derived from observational data. These parameter values are the authoritative defaults used for Solar System UQFF validation and serve as the cross-validation baseline for all heliocentric calculations. The paper includes the exact numerical values, units, physical interpretation, and UQFF equations in which each parameter appears.

UQFF Discovery: Novel application of UQFF calibration constants (κ = 5.0$\times$10-4 day-1, [SSq] =
0.57) uniquely enabling this analysis — establishing a new connection in the UQFF framework not present in Standard Model treatments.

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1. CelestialBody Struct Definition

The CoAnQi `CelestialBody` struct (namespace `CoAnQi::Physics`) encapsulates 12 UQFF parameters per
body:

```cpp
struct CelestialBody {
std::string name;
double Ms; // stellar/body mass [kg]
double Rs; // body radius [m]
double Rb; // buoyancy boundary radius [m]
```

```
double Ts_surface; // surface temperature [K]
double omega_s; // rotation angular frequency [rad/s]
double Bs_avg; // average surface magnetic field [T]
double SCm_density; // SCm density at body surface [kg/m3]
double QUA; // UA charge [C]
double Pcore; // normalized core pressure [Pa]
double PSCm; // normalized SCm pressure [Pa]
double omega_c; // core angular frequency [rad/s]
};
```
``

---

2. Sun

2.1 Canonical Parameters

``cpp

```
CelestialBody sun = {
"Sun",
1.989e30, // Ms: 1.989 \times 103° kg (IAU 2015)
6.96e8, // Rs: 6.96 \times 108 m = 696,000 km
1.496e13, // Rb: 1.496 \times 1013 m ˜ 100 AU (heliosphere)
5778.0, // Ts_surface: 5778 K (photospheric effective temperature)
2.5e-6, // omega_s: 2.5 \murad/s (surface rotation, equatorial ~25 days)
1e-4, // Bs_avg: 100 \muT average dipole field
1e15, // SCm_density: 1015 kg/m3 (UQFF calibrated)
1e-11, // QUA: 10?11 C
1.0, // Pcore: normalized (physical ~2.5 \times 1016 Pa)
1.0, // PSCm: normalized SCm pressure
2.0M_PI/(11.0365.2586400) // omega_c: 2p / (11-year solar cycle)
};
```

2.2 UQFF Outputs at t = 0

Component	Value	Notes
$\mu_s(0)$	$\approx 2.03 \times 10^{22}$ T\cdotm3	Solar dipole moment
$U_{g1}(R_\oplus, 0)$	$\approx 9.26 \times 10^{22}$ N	At Earth orbit
$U_{g2}(R_\oplus, 0)$	$\approx 9.83 \times 10^6$ N	Below buoyancy boundary
$U_m(0)$	$\approx 2.26 \times 10^{16} \cdot (1 - e^{-\gamma t})$	String magnetism
$E_{\text{react}}(0)$	$\approx 8.74 \times 10^{45} \cdot (1 - e^{-\kappa t})$	SCm reactor

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3. Earth

3.1 Canonical Parameterscpp

```
CelestialBody earth = {
"Earth",
5.972e24, // Ms: 5.972 \times 1024 kg
6.371e6, // Rs: 6.371 \times 106 m (mean radius)
1e7, // Rb: 107 m (Van Allen belt inner edge)
288.0, // Ts_surface: 288 K (global mean surface temperature)
```

```
7.292e-5, // omega_s: 7.292 \times 10-5 rad/s (1 sidereal day)
3e-5, // Bs_avg: 30 \muT (Earth's mean field ~50 \muT dipole/2)
1e12, // SCm_density: 1012 kg/m3 (UQFF calibrated for rocky planet)
1e-12, // QUA: 10?12 C
1e-3, // Pcore: normalized (physical ~3.6 \times 1011 Pa inner core)
1e-3, // PSCm: normalized SCm pressure
2.0M_PI/(1.0365.2586400) // omega_c: 2p / (1-year orbital cycle)
};
### 3.2 UQFF Outputs at t = 0
| Component | Value | Notes |
|-----------|-------|-------|
| $\omega_s(0)$ | $7.292 \times 10^{-5}$ rad/s | Rotation rate |
| $U_{g1}(R_{\text{Moon}}, 0)$ | Scaled by $R_\oplus^3 / R_{\text{Moon}}^3$ |
Lunar influence |
| Physical core pressure | $3.6 \times 10^{11}$ Pa | Inner-outer core boundary
|
---
## 4. Jupiter
### 4.1 Canonical Parameterscpp
CelestialBody jupiter = {
"Jupiter",
1.898e27, // Ms: 1.898 \times 1027 kg (1/1047 solar mass)
6.9911e7, // Rs: 6.9911 \times 107 m (equatorial)
1e8, // Rb: 108 m (magnetosphere inner edge)
165.0, // Ts_surface: 165 K (cloud-top effective temperature)
1.76e-4, // omega_s: 1.76 \times 10-4 rad/s (9.93-hour rotation)
4e-4, // Bs_avg: 400 \muT (Jovian dipole ~420 \muT equatorial)
1e13, // SCm_density: 1013 kg/m3 (metallic hydrogen mantle)
1e-11, // QUA: 10?11 C (stormy ionosphere)
1e-3, // Pcore: normalized
1e-3, // PSCm: normalized
2.0M_PI/(11.86365.25*86400) // omega_c: 2p / (11.86-year Jupiter orbital
period)
};
``
```

4.2 UQFF Significance

Jupiter's `omega_c` period (11.86 years) is nearly identical to the Sun's solar cycle period (11 years), suggesting a resonance coupling between the solar SCm oscillation and Jupiter's orbital dynamics — consistent with proposed solar-Jupiter tidal forcing models.

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5. Neptune

5.1 Canonical Parameters

```
``cpp
CelestialBody neptune = {
"Neptune",
1.024e26, // Ms: 1.024 \times 1026 kg
```

```
2.4622e7, // Rs: 2.4622 \times 107 m (equatorial)
5e7, // Rb: 5 \times 107 m (magnetospheric inner boundary)
72.0, // Ts_surface: 72 K (effective temperature)
1.08e-4, // omega_s: 1.08 \times 10-4 rad/s (16.11-hour rotation)
1e-4, // Bs_avg: 100 \muT (highly tilted dipole ~14–16 \muT at 1 R_N)
1e11, // SCm_density: 1011 kg/m3 (ice giant, water/ammonia mantle)
1e-13, // QUA: 10?13 C
1e-3, // Pcore: normalized
1e-3, // PSCm: normalized
2.0M_PI/(164.8365.25*86400) // omega_c: 2p / (164.8-year Neptune orbital period)
};
``
```

5.2 UQFF Significance

Neptune's 164.8-year period (completed its first full orbit since discovery in 2011) represents the outer edge of the Solar System's UQFF coherence zone. Its SCm density is the lowest of the four bodies, consistent with a water-ice mantle having less SCm confinement than gas giants or rocky planets.

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6. Comparative Analysis

Parameter	Sun	Earth	Jupiter	Neptune
M_s (kg)	1.99×10^{30}	5.97×10^{24}	1.90×10^{27}	1.02×10^{26}
R_s (m)	6.96×10^8	6.37×10^6	6.99×10^7	2.46×10^7
T_s (K)	5778	288	165	72
ω_s (rad/s)	2.5×10^{-6}	7.3×10^{-5}	1.76×10^{-4}	1.08×10^{-4}
ρ_{SCm} (kg/m3)	10^{15}	10^{12}	10^{13}	10^{11}
Orbital period	—	1 year	11.86 yr	164.8 yr

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7. Data Sources

All values cross-referenced with:
- IAU 2015 nominal solar/planetary radii and masses
- NASA Planetary Fact Sheets (planetary.org/explore)
- NSSDC Solar System Exploration data
- UQFF calibration: SCm_density and QUA fitted to observed UQFF validation metrics

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8. Conclusion

The four canonical Solar System CelestialBody instances (Sun, Earth, Jupiter, Neptune) provide the operational test suite for all heliocentric UQFF calculations. The 12-parameter struct captures the complete set of UQFF-relevant quantities, from classical (mass, radius, temperature) to

UQFF-specific (SCm_density, QUA, Pcore, PSCm). The Jupiter–solar-cycle resonance and the Neptune UQFF coherence boundary are notable discoveries from this parameterization.

<!-- PKG-AGN-S225 -->

Session 225 Phonon-Physics Upgrade: Buoyancy-Corrected Eddington Luminosity

> *Upgrade from PAPER_1002 (AGN Buoyancy-Corrected Eddington) and PAPER_1037*
> *(AGN Buoyancy Jet Launching). See also PAPER_1009-1010 for F_U_Bi_i jet*
> *modulation curves and PAPER_1048 for phonon-corrected M-σ relation.*

The SCm vacuum buoyancy partially opposes gravitational radiation pressure,
raising the effective Eddington luminosity:

$$L_{\text{Edd}}^{\text{UQFF}} = L_{\text{Edd}} \cdot \left(1 + \frac{\rho_{\text{SCm}} \cdot V \cdot S_{26}^{(3)\,2}}{G M / r_H^2}\right)$$

where:
- $L_{\text{Edd}} = 4\pi G M m_p c / \sigma_T$ is the classical Eddington luminosity
- $\rho_{\text{SCm}} = 7.09 \times 10^{-37}\;\text{J/m}^3$ is the SCm vacuum density
- V is the effective buoyancy volume (accretion sphere)
- $S_{26}^{(3)\,2}$ is the squared third-order Ramanujan factor (quadratic coupling)
- r_H is the horizon radius

Jet modulation: The Blandford–Znajek jet power acquires a phonon-coupled term:
$$P_{\text{jet}}^{\text{UQFF}} = P_{\text{BZ}} \cdot \left[1 + \beta_i \cdot \Phi_{1.25\,\text{THz}} \cdot \left(\frac{B}{B_{\text{crit}}}\right)^2\right]$$

where $\Phi_{1.25\,\text{THz}} = \cos(\omega_{\text{SCm}} \cdot t)$ modulates jet power at the phonon frequency.

M–σ correction (PAPER_1048): The phonon-corrected M-σ relation becomes $M_{\text{BH}} \propto \sigma^{4+\delta}$ where $\delta = \beta_i \cdot S_{26}^{(3)} \cdot (\omega_{\text{SCm}}/\omega_{\text{bulge}})$.

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

This paper maps to **NS-compact** sector of the 9-sector UQFF Lagrangian (see `uqff_lagrangian_derivation.py`).

§A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\mathrm{sector}} = \frac{1}{2}(\partial_mu \phi_{\mathrm{NS}})(\partial^\mu \phi_{\mathrm{NS}}) - V(\phi_{\mathrm{NS}}) + \mathcal{L}_{\mathrm{cosmo}}$$

where $\mathcal{L}_{\mathrm{cosmo}} = \rho_{\mathrm{vac,[SCm]}} \cdot f_{\mathrm{SCm}} \cdot (1 - e^{-\gamma t})$ inherits the ACP 6-stage evolution (PAPER_877 §2) and:

$$V(\phi_{\mathrm{NS}}) = \frac{1}{2} m^2 \phi_{\mathrm{NS}}^2 + \frac{\lambda}{4!} \phi_{\mathrm{NS}}^4 + \kappa \cdot \rho_{\mathrm{vac,[SCm]}} \cdot \phi_{\mathrm{NS}}$$

§A.3 Euler-Lagrange Equation of Motion

$$\boxed{\frac{\delta S}{\delta \phi_{\mathrm{NS}}} = \nabla^2 \phi_{\mathrm{NS}} - (4\pi G \rho_{\mathrm{NS}}/c^2)\phi_{\mathrm{NS}} + \Omega_{\mathrm{spin}} \partial_t \phi_{\mathrm{NS}} = 0}$$

§A.4 Cosmogenesis Linkage Chain

$$\text{PAPER\_877 Axioms} \xrightarrow{\text{DPM + ACP}} \rho_{\mathrm{vac}} = \rho_{\mathrm{UA}} + \rho_{\mathrm{SCm}} \xrightarrow{\text{Stage 5}} U_{b,\mathrm{seed}} \xrightarrow{\text{4 forces}} F_U_Bi_i \xrightarrow{\text{sector E-L}} \delta S/\delta \phi_{\mathrm{NS}} = 0$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four U_g forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

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§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

The canonical VDS ratio $\rho_{\mathrm{vac,[SCm]}} / \rho_{\mathrm{UA}} = 1.894$ governs the double-exponential vacuum condensate profile:

$$\rho_{\mathrm{vac}}(r) = \rho_{\mathrm{vac,[SCm]}} \cdot \exp\!\left(-\exp\!\left(-\frac{r - r_0}{\lambda_{\mathrm{VDS}}}\right)\right)$$

For this system, the local VDS sub-ratio is 0.173 (near-threshold regime), placing it in the $t \to \pi$ collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER_877 cosmogenesis Stage 1 vacuum density initialization: $\rho_{\mathrm{vac}} = \rho_{\mathrm{UA}} + \rho_{\mathrm{SCm}} = 7.799 \times 10^{-36}$ kg/m3.

§B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\mathrm{DVP}} = 17, \quad n_{\mathrm{channel}} = 5/26$$

Since $p_{\mathrm{DVP}} = 17$ is **sub-threshold** (threshold at $p > 26$), the system's vacuum topology inherits sub-threshold damping from the DVP lattice, producing smooth rather than resonant UQFF coupling profiles. The DVP framework traces to PAPER_877 proto-nuclear shell formation: the DPM proportion pair $(f_{\mathrm{UA}}' + f_{\mathrm{SCm}} = 1)$ constrains which primes are accessible at each atomic number.

§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is **104 yr** (spin-down equilibrium):

$$\mathcal{F}_{\mathrm{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_U_b \cdot \left(1 - e^{-[SSq] \cdot m/M_\odot}\right) \cdot \cos\!\left(\frac{2\pi j}{26}\right)$$

The $\tanh$ saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\mathrm{BSH,sat}} = \mathcal{F}_{\mathrm{BSH}} \cdot \left(1 - \tanh\!\left(\frac{t - t_{\mathrm{sat}}}{\tau_{\mathrm{BSH}}}\right)\right)$$

connecting to the PAPER_877 Stage 5 buoyancy seed $U_{b,\mathrm{seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\mathrm{SCm}}$ which initializes the harmonic series at cosmogenesis.

§B.4 Production-Scale Consistency

Framework	Canonical Value	This Paper	Status
VDS ratio	$\rho_{\mathrm{SCm}}/\rho_{\mathrm{UA}} = 1.894$	Local sub-ratio = 0.173	PASS Threshold-consistent
DVP prime	$p_k \in$ {2,3,...,113}	$p_{\mathrm{DVP}} = 17$	PASS Sub-threshold
BSH layers	26 harmonic terms	j = 1...26, $\cos(2\pi j/26)$	PASS Full 26D projection
κ decay	5.0×10^{-4} day-1	Applied in VDS exponential	PASS Canonical
[SSq]	0.57	Applied in BSH saturation	PASS Canonical

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§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Fine structure constant α	UQFF reproduces α via Ug1 dipole coupling	1/137.036	PDG 2024	PASS Consistent
Cosmological constant Λ	1.1$\times$10-52 m-2 (UQFF vacuum term)	1.114$\times$10-52 m-2	Planck 2018	PASS Consistent
Proton decay rate	κ = 0.0005/day $\to$ Γ_p suppression	< 4.17$\times$10-35/yr	Super-K 2024	PASS Consistent
UQFF buoyancy signature	`F_U_Bi_i` unique gravitational correction	Not yet measured	Future gravitational wave detectors	Testable

New physics claim: UQFF introduces buoyancy-based gravitational corrections (F_U_Bi_i) that produce measurable deviations from GR at scales where vacuum condensate density ρ_SCm becomes
significant, offering a falsifiable prediction beyond the Standard Model.

Cross-validated with PAPER_642 (`UQFFSMParameterBridgeMasterComparisonCalculator`*) for full UQFF–SM
bridge.*

References

- Source: grok_{share_381a8f}.txt lines 3200–4100 (inline CelestialBody defaults)
- Related: PAPER_170 (CelestialBody 12-Field Parameter Space), PAPER_182 (Variable Reference)
- CP2 Class: `CoAnQiSolarSystemReferenceCalculator`

Appendix: Session 225 Cross-References (PAPER_1000–1081)

> *Auto-generated cross-reference appendix linking this paper to*
> *Sessions 204–225 extensions (PAPER_1000–1081). Added by*
> `update_corpus_crossrefs.py` *(Session 225, April 2026).*

Paper	Title
PAPER_1002	AGN Buoyancy-Corrected Eddington Luminosity
PAPER_1009	3C273 AGN F_U_Bi_i Jet Modulation
PAPER_1010	TON618 AGN F_U_Bi_i Jet Modulation
PAPER_1037	AGN Buoyancy Jet Calculator — SCm Jet Launching
PAPER_1048	M-Sigma Phonon-Corrected Relation
PAPER_1041	SCm Cool-Core Buoyancy Balance AGN Feedback
PAPER_1079	Galaxy Cluster Cooling-Flow Buoyancy Suppression
PAPER_1029	Barocentric Earth Orbital Buoyancy
PAPER_1043	F_U_Bi_i Multi-System Buoyancy Curve Sweep
PAPER_1072	SCm Activation Function Phonon Threshold
PAPER_1073	SCm Phonon-Driven Inflation Vacuum Buoyancy
PAPER_1065	Buoyancy Lagrangian EOM Variational Derivation

12 cross-reference(s) identified.

Appendix: Session 204 Codebase Upgrade Reference

> *Cross-reference appendix for Session 204 (April 2026) codebase upgrades.*
> *Added by* `upgrade_kozima_ramanujan_appendices.py`*. For detailed derivations,*
> *see PAPER_840/851/852/855.*

S204.1 Kozima-UQFF LENR Integration

Module	Purpose	Key Result
`fneutron_s26_coupling.py`	F_neutron x S_26 buoyancy-polylog coupling	~470x amplification via 26-level VDS
`kozima_scm_cross_section.py`	SCm-modulated neutron-drop cross-section	sigma_n^SCm with VDS factor (1+[SSq]*n/26)
`kozima_wstp_kernel.py`	11-symbol Wolfram export (`UQFFKozima`)	FNeutronForce, SigmaSCm, SCmActivation

Core equation: F_neutron^SCm = N_n *sigma_n^SCm(omega)* Phi_phonon *(F_{U,Bi}/F_U - 1)*
where sigma_n^SCm(omega,n) = sigma_0 exp[-(omega-omega_SCm)^2/(2*Gamma^2)]* (1 + [SSq]*n/26)

S204.2 Ramanujan 26-State Summation

Module	Purpose	Key Result
`ramanujan_polylog_s26.py`	Li_26([SSq]) via Euler-Ramanujan acceleration	15.7+ digits in 53 terms
`s26_wstp_kernel.py`	8-symbol Wolfram export (`UQFFS26`)	S26, R26, NaiveLi, S26VDS

Core equation: S_26(z) = Li_26(z) = eta_26(z)/(1-2^{1-26}) + 2^{1-26}/(1-2^{1-26}) * Li_26(z^2)

S204.3 Mock Theta Functions (26-State)

Module	Purpose	Key Result
`mock_theta_q26.py`	f_26(q), phi_26(q), psi_26(q) q-series	Proper q-Pochhammer (a;q)_n

Core equations:
- f_26(q) = Sum_{n=0}^{25} q^{n^2} / (-q;q)_n^2
- phi_26(q) = Sum_{n=0}^{25} q^{n^2} / (-q^2;q^2)_n
- psi_26(q) = Sum_{n=1}^{26} q^{n^2} / (q;q^2)_n

S204.4 Ramanujan 1/pi with UQFF Modification

Module	Purpose	Key Result
`ramanujan_pi_uqff.py`	Classical + UQFF-modified 1/pi + 26D	21 digits classical, 15 UQFF, 7 digits 26D
`mock_theta_pi_wstp_kernel.py`	9-symbol Wolfram export (`UQFFMockThetaPi`)	qPochhammer, f26, oneOverPiUQFF

Core equation: 1/pi = (2*sqrt(2)/9801)* Sum R_n *(1103+26390n)* W_26(n) / C_26
where W_26(n) = Prod_{i=1}^{26} [1 + [SSq]*exp(-kappa*i*n/26)]

S204.5 Calibration Constants (Canonical)

Symbol	Value	Description
[SSq]	0.57	Universal Quantized Factor
kappa	5.787 x 10^-9 s^-1	UQFF exponential decay rate
beta_i	0.603	Buoyancy coupling coefficient
H_SCm	0.99	SCm manifold completeness
rho_SCm	7.09 x 10^-37 kg/m^3	SCm vacuum density
rho_UA	7.09 x 10^-36 kg/m^3	UA aether vacuum density
omega_SCm	2*pi x 1.25 THz	SCm phonon resonance
sigma_0	10^-4	Base neutron cross-section

Implementation: all modules operational in `CondensedPhysics.py`*,* `CondensedPhysics2.py`*,* `MAIN_{1\_CoAnQi}.cpp`*, and Wolfram kernels (*`uqff_kozima_kernel.wl`*,* `uqff_s26_kernel.wl`*,* `uqff_mock_theta_pi_kernel.wl`*).*

Key References with arXiv/DOI Identifiers

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